

Simultaneous Segmentation of Ventricles and Normal/Abnormal White Matter Hyperintensities in Clinical MRI using Deep Learning

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Abstract

Multiple sclerosis (MS) diagnosis and monitoring rely heavily on accurate assessment of brain MRI biomarkers, particularly white matter hyperintensities (WMHs) and ventricular changes. Current segmentation approaches suffer from several limitations: they typically segment these structures independently despite their pathophysiological relationship, struggle to differentiate between normal and pathological hyperintensities, and are poorly optimized for anisotropic clinical MRI data. We propose a novel 2D pix2pix-based deep learning framework for simultaneous segmentation of ventricles and WMHs with the unique capability to distinguish between normal periventricular hyperintensities and pathological MS lesions. Our method was developed on FLAIR MRI scans from 300 MS patients and validated across both local and public datasets (MSSEG2016). Compared to established methods (SynthSeg, Atlas Matching, BIANCA, LST-LPA, LST-LGA, and WMH-SynthSeg), our approach achieved superior performance for both ventricle segmentation (Dice: 0.801 ± 0.025 , HD95: 18.46 ± 7.1 mm) and WMH segmentation (Dice: 0.624 ± 0.061 , precision: 0.755 ± 0.161). Furthermore, our method successfully differentiated between normal and abnormal hyperintensities with a Dice coefficient of 0.647. Cross-dataset validation demonstrated consistent performance for ventricle segmentation while revealing important generalizability insights for WMH segmentation. Notably, our approach demonstrated exceptional computational efficiency, completing end-to-end processing in approximately 4 seconds per case—up to 36 times faster than baseline methods—while maintaining minimal resource requirements. This combination of improved accuracy, clinically relevant differentiation capability, and computational efficiency addresses critical limitations in current neuroimaging analysis, potentially enabling integration into routine clinical workflows and enhancing MS diagnosis and monitoring.

Keywords: Brain MRI Segmentation; pix2pix Deep Learning; White Matter Hyperintensities (WMH) and Ventricular Segmentation; White Matter Hyperintensities (WMH) Differentiation

Key Points:

1. Novel Simultaneous Segmentation with Clinical Differentiation

We developed the first deep learning framework to simultaneously segment ventricles and white matter hyperintensities (WMH) while uniquely distinguishing between normal periventricular hyperintensities and pathological MS lesions—a critical differentiation that prevents overestimation of disease burden in clinical practice.

2. Superior Performance Optimized for Clinical Reality

Our pix2pix-based approach achieved state-of-the-art accuracy (Dice: 0.801 for ventricles, 0.624 for WMH) while being specifically designed for anisotropic clinical MRI data that most research methods cannot handle effectively, bridging the gap between research innovations and routine clinical imaging protocols.

3. Exceptional Computational Efficiency Enabling Clinical Integration

The method processes complete cases in approximately 4 seconds—up to 36 times faster than existing approaches—with minimal resource requirements (15% CPU, ~1 GB RAM), making real-time analysis feasible during clinical sessions and accessible across diverse healthcare settings including resource-limited environments.

1. Introduction

Multiple Sclerosis (MS) is a chronic, inflammatory, demyelinating disease of the central nervous system affecting approximately 2.8 million people worldwide (Walton et al., 2020). Diagnosis, monitoring, and treatment planning for MS heavily rely on accurate assessment of brain structural changes visible on Magnetic Resonance Imaging (MRI), particularly white matter hyperintensities (WMHs) and ventricular alterations (Tran et al., 2022). These imaging biomarkers are crucial for evaluating disease progression, treatment efficacy, and prognosis.

White matter hyperintensities appear as bright regions on T2-weighted and fluid-attenuated inversion recovery (FLAIR) MRI sequences, representing areas of demyelination, inflammation, and tissue damage (Melazzini et al., 2021). Their spatial distribution, count, and volumetric progression correlate with clinical disability measures and cognitive impairment (Wang et al., 2022). Furthermore, the burden and evolution patterns of these hyperintensities have been associated with both disease progression and therapeutic response (Melazzini et al., 2021). Historical studies have established WMH as independent predictors of cognitive decline and functional outcomes across various neurological conditions (DeCarli et al., 2005). Ventricular enlargement serves as an important indicator of brain atrophy, another hallmark of MS progression (Laso et al., 2023; McKinley et al., 2021).

Current clinical practice relies heavily on qualitative visual assessment by radiologists who manually identify and characterize lesions across multiple MRI slices (Tran et al., 2022). This process is time-consuming, labor-intensive, and subject to significant inter-observer variability, potentially leading to inconsistent diagnosis and treatment decisions (Park et al., 2021). The complex nature of MS pathology requires precise differentiation between pathological hyperintensities and normal appearing white matter signal variations, a distinction challenging even for experienced specialists (Rakić et al., 2021).

Despite advances in neuroimaging analysis, several critical limitations persist in automated segmentation tools for neurodegenerative diseases. First, almost no current approach simultaneously addresses ventricle and WMH segmentation within a unified framework, despite their anatomical and pathophysiological relationship (Atlason et al., 2022). Umirzakova et al., (2025) demonstrated that ventricle morphology and proximity significantly influence the

interpretation of periventricular hyperintensities, yet most segmentation approaches treat these structures independently. This disconnected approach fails to capitalize on contextual information that could improve segmentation accuracy.

Second, current methods struggle to differentiate between normal hyperintensities resulting from cerebrospinal fluid (CSF) contamination and pathological lesions. This distinction is crucial for accurate disease quantification, as Griffanti et al. (2018) and Dadar et al. (2021) emphasized that misclassification of normal appearing white matter signal variations as lesions can lead to overestimation of disease burden. McKinley et al. (2021) demonstrated that up to 30% of automatically detected hyperintensities in routine clinical scans may represent normal anatomical variants rather than pathology.

Third, the mismatch between research methodologies and clinical data characteristics presents a significant obstacle. Most state-of-the-art segmentation methods assume isotropic 3D volumes with high resolution in all dimensions (Billot et al., 2023; Atlason et al., 2022; La Rosa et al., 2020; Raut et al., 2024; Rondinella et al., 2024; Ulloa-Poblete et al., 2023; Laso et al., 2023; Li et al., 2024; Cathala et al., 2025). However, routine clinical MRI protocols typically produce anisotropic data with significant slice thickness, creating a fundamental incompatibility. Dong et al. (2022) highlighted how this anisotropy presents substantial challenges for conventional 3D segmentation approaches, which may either fail entirely or require computationally intensive preprocessing steps. Recent work by Tran et al. (2022) further demonstrated that inconsistencies in imaging protocols and resolution contribute significantly to variability in WMH quantification, particularly affecting the detection of small lesions and periventricular boundaries.

The computational demands of 3D deep learning models present another practical barrier to clinical implementation. While these models have demonstrated impressive performance in research settings, Tran et al. (2021) found that their deployment in typical clinical environments is constrained by hardware limitations and workflow integration challenges. This computational burden is particularly problematic in resource-limited healthcare settings where advanced computing infrastructure may not be available.

To address these limitations, we propose a novel 2D pix2pix deep learning framework specifically designed for the simultaneous segmentation of ventricles and white matter

hyperintensities in anisotropic clinical MRI data. Our approach builds upon recent innovations in generative adversarial networks for medical image segmentation that have demonstrated improved handling of structural complexity and boundary precision in neuroimaging applications (Zeng et al., 2020; Zhang et al., 2022). Building upon our previous work on pix2pix-based gray matter segmentation for WMH categorization (Bawil et al., 2024), our current approach extends this methodology to simultaneously segment multiple brain structures critical for MS evaluation. Our method uniquely distinguishes between normal periventricular hyperintensities and pathological lesions, offering a comprehensive solution compatible with routine clinical workflows. The pix2pix architecture (Isola et al., 2016) serves as the foundation of our approach, modified to accommodate our multi-class segmentation task.

Unlike methods requiring resource-intensive 3D processing, our slice-based approach offers significant advantages for clinical implementation. As highlighted by Hossain et al. (2024) and Hashemi et al. (2022), 2D slice-based methods can achieve comparable accuracy to 3D approaches while dramatically reducing computational requirements, particularly important for handling anisotropic clinical scans. Our proposed model segments four distinct classes: background, ventricles, normal WMH (CSF-contaminated hyperintensities), and abnormal WMH (pathological lesions). This multi-class approach represents a significant advancement over existing tools that typically segment either ventricles or WMH independently.

A key strength of our work is the substantial dataset comprising 300 clinical MRI scans with comprehensive expert annotations of ventricles and both types of WMH. Our model incorporates FLAIR sequence as an input, enabling more accurate differentiation between periventricular CSF-contaminated hyperintensities and true pathological lesions (Griffanti et al., 2016; Zhu et al., 2022). Our method is designed with clinical utility as a primary goal, focusing on compatibility with standard clinical MRI protocols without requiring specialized acquisition parameters or time-consuming preprocessing. This practical approach significantly distinguishes our work from many research-oriented segmentation methods that perform well on research-grade datasets but fail to translate effectively to routine clinical data.

This article continues with detailed descriptions of our dataset and pix2pix segmentation methodology, evaluation strategy, results with quantitative and qualitative analyses, followed by a discussion of clinical implications and limitations.

2. Materials and Methods

2.1. Study Population

This study analyzed data from 300 MS patients imaged at Golgasht Medical Imaging Center, Tabriz, Iran. The cohort comprised 79 males (aged 18-57 years, mean = 35.8, SD = 9.3) and 221 females (aged 18-68 years, mean = 37.8, SD = 9.7). The study protocol received approval from the Tabriz University of Medical Sciences Research Ethics Committee (IR.TBZMED.REC.1402.902), and all participants provided written informed consent.

2.2. MRI Acquisition Protocol

All imaging was performed on a 1.5-Tesla TOSHIBA Vantage scanner (Canon Medical Systems, Japan). The T2-FLAIR images were acquired with the following parameters: TR = 10000 ms, TE = 100 ms, TI = 2500 ms, flip angle = 90°, FOV = 230 × 230 mm², acquisition matrix = [0, 256, 192, 0], resulting in non-isotropic data with voxel dimensions of (0.9, 0.9, 6) millimeters. This substantial difference between in-plane resolution and slice thickness informed our decision to implement a 2D rather than 3D segmentation approach.

2.3. Ground Truth Development

Manual segmentation was performed on all 300 MS patients' FLAIR images by a neuroradiologist with over 20 years of expertise in neuroimaging. A custom Python-based GUI facilitated the segmentation process. Four distinct classes were delineated: (1) background, (2) ventricles, (3) normal white matter hyperintensities (WMH), and (4) abnormal WMH.

The segmentation methodology followed these steps:

1. Ventricle Segmentation: Initial manual delineation of ventricular boundaries, followed by statistical morphological post-processing for refinement and final manual verification.
2. Abnormal WMH Segmentation: Identification of hyperintense regions representing MS lesions with subsequent processing and expert verification, as same as step 1.

3. Normal WMH Identification: Creating a periventricular boundary by dilating ventricle masks, identifying high-intensity pixels within this boundary, excluding regions overlapping with abnormal WMH, and manual refinement focusing on frontal and occipital horn vicinities.

This approach ensured differentiation between normal hyperintensities (related to CSF contamination) and abnormal hyperintensities (true candidates of MS lesions), with the ventricular system coded in blue, abnormal WMH in red, and normal WMH in green, as shown in Figure 1.

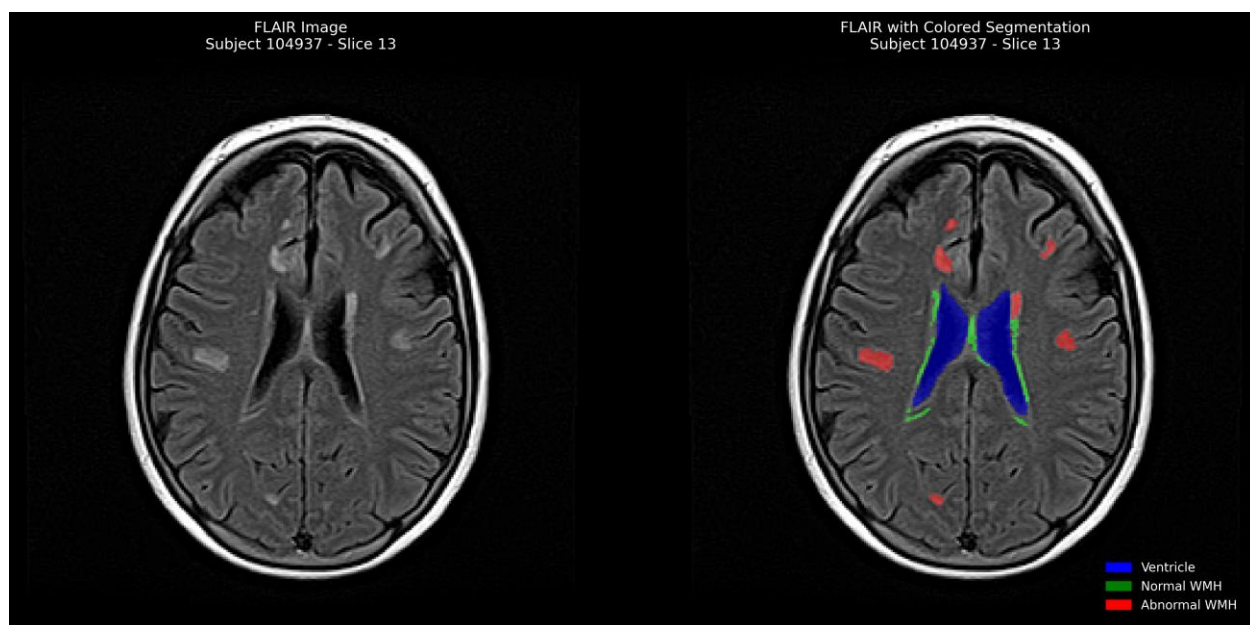


Figure 1. A sample image of MRI FLAIR data with corresponding manual segmentation and annotation. The left image is an original image of a MS patient. The left image shows abnormal WMH, normal WMH, and ventricles by red, green, and blue colors, respectively.

2.4. Public Dataset

To enhance the generalizability and robustness of our findings, we incorporated the publicly available MSSEG 2016 challenge dataset (Commowick et al., 2018). This dataset comprises 15 MS patients' MRI data acquired from 3 different imaging centers with standardized acquisition protocols. The original MSSEG 2016 dataset provides expert-annotated ground truth segmentations only for white matter hyperintensities (MS lesions). Since our methodology requires simultaneous segmentation of both ventricles and WMHs, we supplemented this dataset

by having our neuroradiologist expert manually segment the ventricular regions using the same protocol described in section 2.3. This comprehensive annotation enables evaluation of our proposed multi-class segmentation approach across diverse imaging conditions and patient populations.

2.5. Pre-processing Pipeline

Our pre-processing pipeline forms a critical foundation for successful application of deep learning for brain MRI segmentation, as illustrated in Figure 2. This pipeline standardizes input data, reduces noise, and prepares images for optimal model training and inference.

2.5.1. Noise Reduction

MR images inherently contain various noise artifacts that impede effective segmentation. We employ a two-stage noise reduction strategy that applies a median filter (kernel size = 3×3) to suppress salt-and-pepper noise while preserving edges, followed by selective application of a Gaussian filter ($\sigma = 1.0$) to smooth remaining artifacts while retaining structural details. We deliberately apply minimal preprocessing, as our deep learning architecture is designed to handle more sophisticated noise patterns, including bias field inhomogeneity, directly during training following evidence that deep convolutional networks can learn to distinguish signal from noise given sufficient training examples (Wu et al., 2024).

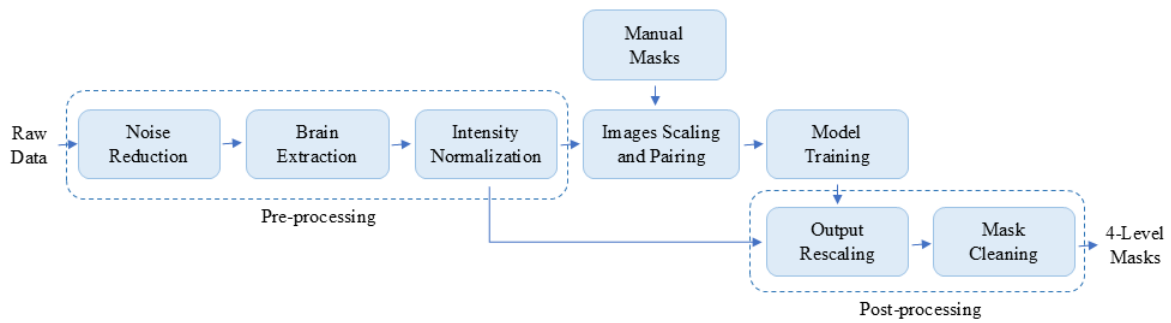


Figure 2. Block diagram of our proposed method.

2.5.2. Brain Extraction

Our framework employs a morphology-based approach for brain extraction that prioritizes computational efficiency while providing sufficiently accurate boundaries for subsequent processing. The process begins with thresholding based on intensity values to separate brain from background, followed by morphological operations to remove small isolated regions. We then generate an elliptical mask approximating the brain contour using the second-order moments of the binary image, and refine it through contour analysis to better conform to actual brain boundaries. This approach yields a binary brain mask that identifies the general brain region without requiring complex algorithms or excessive computational resources. The resulting mask provides sufficient information for subsequent scaling operations while eliminating most non-brain tissues.

2.5.3. Intensity Normalization

We implement a slice-based intensity normalization technique addressing both inter- and intra-patient intensity variations. Unlike conventional global approaches, our method establishes parameters independently for each image slice. The minimum intensity value derives from average background intensity specific to each slice, while the maximum intensity value is determined by analyzing peripheral structures (scalp, skull, and surrounding tissues) obtained by subtracting the brain mask from the full image. This approach ensures intra-patient consistency since background and skull structures remain relatively constant across slices from the same patient, while providing inter-patient adaptability by accommodating anatomical variations. Recent work by Huang et al. (2022) has demonstrated that such adaptive intensity normalization strategies significantly improve segmentation performance, particularly for detecting subtle intensity variations characteristic of MS lesions. Intensity values are scaled to the range [0,1] using Eq. 1, where normalization parameters are derived using our slice-specific methodology.

$$I_{normalized} = \frac{I - I_{min}}{I_{max} - I_{min}} \quad (1)$$

2.5.4. Paired-image Generation

The final pre-processing step prepares standardized input for our conditional Generative Adversarial Network (cGAN) architecture, the pix2pix model. We scale the brain region to maximize coverage within the image frame using affine transformations derived from the brain mask's geometric properties, then apply identical transformations to the corresponding ground truth segmentation mask to maintain spatial alignment. The processed image and its ground truth mask are horizontally concatenated to create a paired 256×512 pixel composite image compatible with the pix2pix architecture (Isola et al., 2016). This paired-image approach enables the model to learn the mapping between MR images and their segmentation masks within a unified spatial context, while standardized scaling ensures brain structures occupy maximum possible area within the image, reducing the proportion of zero-value pixels that can adversely affect gradient updates during training.

2.6. Network Architecture and Implementation

Our method employs a cGAN architecture—specifically the pix2pix model—as the core component for brain segmentation. As illustrated in Figure 2, this model bridges the pre-processing stages and post-processing refinements in our pipeline. The detailed network architecture and implementation strategy are described in the following and visually represented in Figure 3.

2.6.1. Pix2Pix Framework

We leverage the pix2pix framework for its proven effectiveness in image-to-image translation while preserving structural coherence. The framework consists of two networks: a generator (modified U-Net) that translates MR images into segmentation masks and a discriminator (PatchGAN) that evaluates mask realism. As depicted in Figure 3, the generator receives the pre-processed MR images and produces segmentation masks, while the discriminator simultaneously learns to distinguish between generated masks and ground truth annotations. This adversarial process drives the generator to produce increasingly accurate segmentations that can fool the discriminator.

2.6.2. U-Net Backbone

The generator employs a modified U-Net architecture (Ronneberger et al., 2015) with an encoder-decoder structure and skip connections. The encoder comprises eight downsampling blocks (convolutional layer, batch normalization, leaky ReLU), while the decoder includes seven upsampling blocks (transposed convolutions, batch normalization, ReLU, selective dropout).

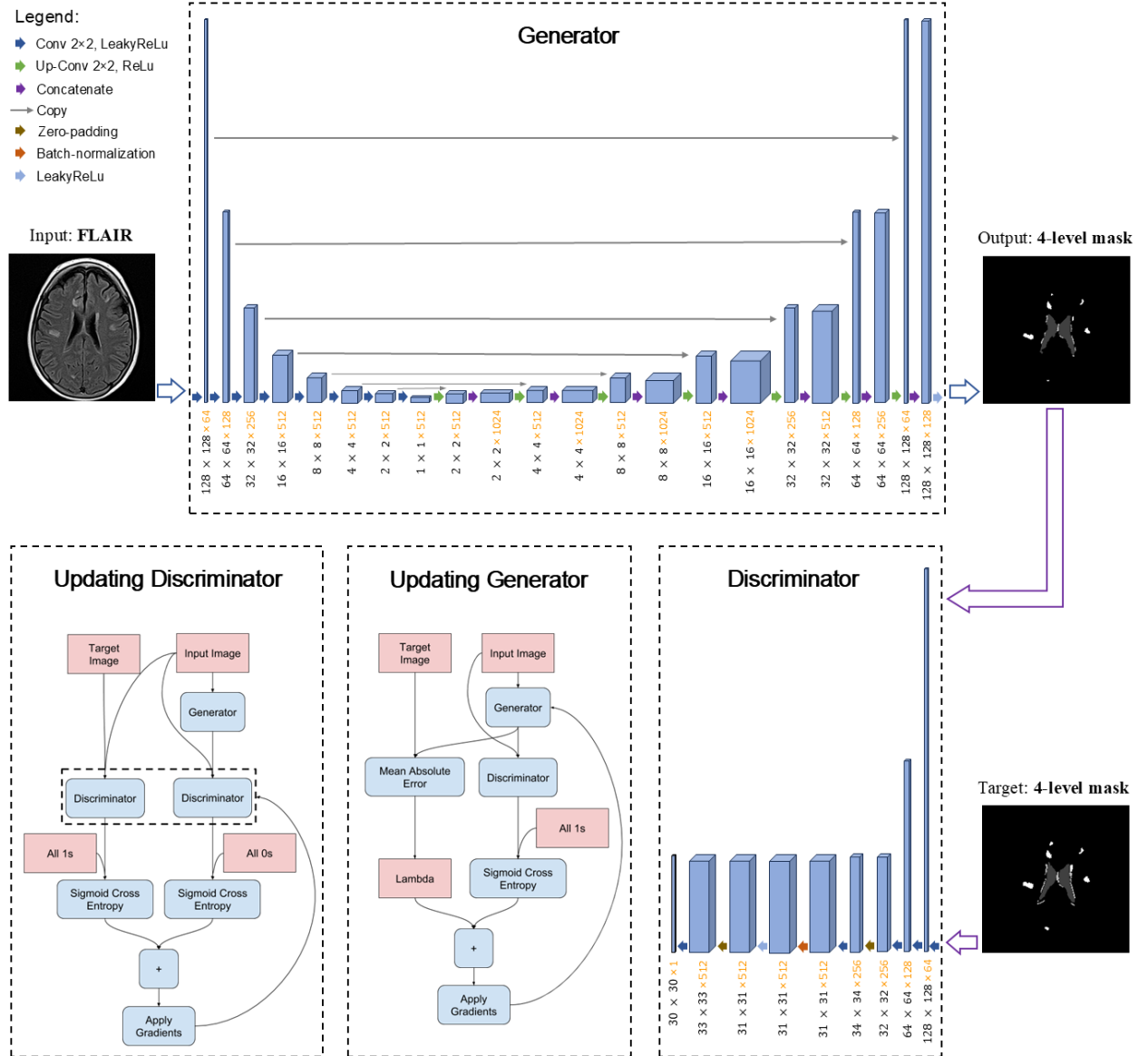


Figure 3. Pix2pix model architecture overview. This system is trained using paired images consisting of input FLAIR sequences and corresponding 4-level segmentation masks as targets. The updating blocks displayed along the bottom row illustrate the learning progression of the respective network components.

The network maintains a channel depth progression of [64, 128, 256, 512, 512, 512, 512, 512] in the encoder path and the reverse in the decoder, ensuring sufficient capacity for feature capture while maintaining computational efficiency. This architecture design aligns with recent advances in GAN-based medical image segmentation that emphasize the importance of multi-level feature extraction for handling complex anatomical boundaries (Raut et al. 2024; Bawil et al., 2024; Isola et al., 2016; Wu et al., 2024).

2.6.3. Training and Testing Strategy

We train our model with a batch size of 1, for 50 epochs, using the Adam optimizer ($\beta_1 = 0.5$, $\beta_2 = 0.999$) with a constant learning rate of 0.0002. The loss function combines adversarial loss (binary cross-entropy) and L1 loss (mean absolute error) defined in Eq. 2.

$$\mathcal{L} = \lambda_{adv} \cdot \mathcal{L}_{adv} + \lambda_{L1} \cdot \mathcal{L}_{L1}, \text{ where } \lambda_{adv} = 1 \text{ and } \lambda_{L1} = 100. \quad (2)$$

This combination encourages both structural accuracy and realistic segmentation boundaries. Training is conducted on paired images as described in Section 4.4, with the generator and discriminator updated alternately according to the standard GAN training procedure illustrated in Figure 3.

For robust evaluation, we split the local dataset into training (70%) and testing (30%) sets using patient-level stratified sampling to prevent data leakage. We rely on the default parameters of the pix2pix framework, which proved effective without extensive hyperparameter tuning, reducing overfitting risk. No cross-validation is performed as the hyperparameters are tuned beforehand and the test set is sufficiently large to provide reliable performance estimates.

Given the limited size of the MSSEG 2016 dataset, 3 patients are randomly selected for fine-tuning the pre-trained model from the local dataset, while the remaining 12 patients serve as an independent test set for cross-dataset evaluation. The final model is selected based on the lowest validation loss achieved during training or fine-tuning.

2.6.4. Post-processing

The raw network output undergoes three critical post-processing steps. First, the predicted segmentation masks are transformed back to the original image space using the inverse of the

scaling transformations applied during pre-processing. This ensures alignment with the original MR images and enables direct comparison with ground truth annotations. Second, the single-channel grayscale output is converted into four separate probability masks representing our four segmentation classes. This conversion uses a distance-based softmax approach where pixel intensities are compared to predefined class values (0.0, 0.25, 0.75, and 1.0), corresponding to background, ventricles, normal white matter, and white matter hyperintensities, respectively. Third, to improve the quality of segmentation results, we apply a series of morphological operations to the predicted masks such as connected component analysis, hole filling, and edge smoothing. These minimal post-processing steps preserve the network's predictions while removing obvious artifacts, resulting in high-precision segmentation of target brain structures with anatomical plausibility.

2.7. Baseline Methods

To evaluate the performance of our proposed approach, we compare it against established methods for both ventricle and hyperintensity segmentation. Since no existing method simultaneously performs four-class segmentation (background, ventricles, normal WMH, abnormal WMH), we conducted separate comparisons for ventricle and hyperintensity segmentation.

2.7.1. Ventricle Segmentation Baselines

SynthSeg: We employ the SynthSeg model, a deep learning-based approach trained on synthetic data for robust brain MRI segmentation (Billot et al., 2023). Following the official implementation guidelines, we apply SynthSeg to our FLAIR images and extracted ventricle masks from the full brain parcellation output based on their corresponding label values.

Atlas Matching: We implement an atlas-based approach using the MNI152 standard space template (Fonov et al., 2011). The process involves registering each subject's FLAIR image to MNI152 space using FSL's linear registration tool (FLIRT), applying the ventricle atlas mask, performing basic morphological post-processing to refine boundaries, and finally transforming the resulting segmentation back to the subject's native space.

2.7.2. Hyperintensity Segmentation Baselines

BIANCA (Brain Intensity AbNormality Classification Algorithm): We utilize this FSL-based supervised method for WMH segmentation (Griffanti et al., 2016). Following the recommended protocol, BIANCA is trained on our manual segmentations using the same training dataset split as our proposed method to ensure fair comparison.

LST-LPA (Lesion Segmentation Tool - Lesion Prediction Algorithm): This unsupervised lesion segmentation algorithm is implemented under the SPM toolbox (Schmidt et al., 2019). We apply LST-LPA directly to our FLAIR images without parameter modification, as recommended in the official documentation. The output binary masks are obtained in the subject's native space.

LST-LGA (Lesion Segmentation Tool - Lesion Growth Algorithm): This algorithm, also part of the SPM toolbox, requires both FLAIR and T1-weighted images (Schmidt et al., 2012). We pre-register T1 images to the corresponding FLAIR space using FSL's FLIRT command and apply LST-LGA with default parameter settings. Binary output masks are generated in the subject's native space.

WMH-SynthSeg: This extension of the SynthSeg framework specifically addresses white matter hyperintensity segmentation (Laso et al., 2023). Following the official guidelines, we apply WMH-SynthSeg to our FLAIR images and extract the WMH masks based on the designated label in the output.

2.7.3. Adaptation for Fair Comparison

For comparing our approach with baseline methods, several adaptations are necessary. Since our method distinguishes between normal and abnormal WMHs while baseline methods produce only a single WMH class, we use abnormal WMH predictions for direct comparison with the baseline hyperintensity segmentation methods. Similarly, as our model simultaneously segments both ventricles and WMHs, we extract the ventricle class predictions for comparison with ventricle-specific baselines. All evaluations are performed in the subject's native space to avoid registration-induced errors in the quantitative comparisons.

2.8. Evaluation Metrics and Analysis

To comprehensively assess the performance of our proposed method and compare it with the baseline methods, we employ multiple complementary evaluation metrics. These metrics are calculated separately for ventricle segmentation and hyperintensity segmentation.

2.8.1. Detection-based Metrics

Precision: Also known as positive predictive value, precision measures the proportion of correctly identified positive voxels among all voxels classified as positive:

$$Precision = \frac{TP}{TP+FP} \quad (3)$$

where TP represents true positives and FP represents false positives.

Recall: Also known as sensitivity or true positive rate, recall measures the proportion of actual positive voxels that were correctly identified:

$$Recall = \frac{TP}{TP+FN} \quad (4)$$

where FN represents false negatives.

2.8.2. Overlap Metrics

Dice Similarity Coefficient (DSC): The Dice coefficient quantifies the spatial overlap between two segmentations and is defined as:

$$DSC = \frac{2 |A \cap B|}{|A| + |B|} \quad (5)$$

where A represents the predicted segmentation and B represents the ground truth segmentation. DSC ranges from 0 (no overlap) to 1 (perfect overlap).

Jaccard Index (JI): Also known as the Intersection over Union (IoU), the Jaccard index is calculated as:

$$JI = \frac{|A \cap B|}{|A \cup B|} \quad (6)$$

JI also ranges from 0 to 1 and is mathematically related to the Dice coefficient.

2.8.3. Boundary Metrics

Hausdorff Distance: While the traditional Hausdorff distance measures the maximum distance between the boundaries of two segmentations, it is highly sensitive to outliers. Therefore, we used the 95th percentile Hausdorff distance (HD95), which is more robust to small segmentation errors:

$$HD95(A, B) = \max(h_{95}(A, B), h_{95}(B, A)) \quad (7)$$

where $h_{95}(A, B)$ is the 95th percentile of the distances from points in boundary A to their closest points in boundary B . HD95 is measured in millimeters, with lower values indicating better boundary agreement.

2.8.4. Discriminative Metrics

Precision-Recall (PR) Curves: We generate precision-recall curves, which plot precision against recall at different threshold values. PR curve and its area under the curve (AUC-PR), also known as average precision (AP), are particularly informative rather than Receiver Operating Characteristic (ROC) curves for imbalanced datasets such as medical image segmentation where the foreground class is much smaller than the background.

2.8.5. Performance Evaluation

To evaluate the performance of normal versus abnormal WMH classification, we compute a confusion matrix at the lesion level. This approach allows for assessment of how well the algorithm distinguishes between the two types of hyperintensities as distinct entities. Moreover, to evaluate the performance of abnormal WMH and ventricles versus background classification, we compute another confusion matrix.

2.9. Development Environment and Technical Implementation

The deep learning framework was implemented using Python 3.9 with TensorFlow, OpenCV, and scikit-image for image preprocessing, and NumPy and Pandas for data handling and statistical analyses. All experiments ran on a workstation with an Intel Core i7-7700K CPU, 64 GB RAM, and NVIDIA RTX 3060 GPU. The codebase with documentation, preprocessing pipelines, model architectures, training scripts, and evaluation modules is available on GitHub (<https://github.com/Mahdi-Bashiri/Sim-Vent-WMH-Seg>) under an MIT license, including pre-trained model weights for immediate inference.

3. Results

3.1. Model Training and Convergence

The cGAN architecture demonstrated efficient training characteristics and robust convergence behavior across all segmentation tasks. Training progression analysis revealed consistent improvement patterns for all three target structures (ventricles, normal WMH, and abnormal WMH).

3.1.1. Training Progression

The model exhibited systematic convergence across 50 epochs, with notable performance stabilization observed after the tenth epoch. Figure 4 illustrates the progressive improvement in discriminative capability and reduction in segmentation errors throughout the training process. Specifically, the convergence behavior showed distinctive patterns for each target structure:

Ventricle segmentation: The model rapidly established strong performance for ventricle delineation, achieving an AUC-PR of 0.8 by epoch 10 and exceeding 0.85 by epoch 15. This swift convergence can be attributed to the relatively consistent appearance and well-defined boundaries of ventricular structures across the patient cohort.

Abnormal WMH segmentation: A more gradual convergence trajectory achieved an AUC-PR of 0.4 by epoch 6 and stabilized over 0.6 after epoch 16. This pattern reflects the heterogeneous nature of MS lesions with respect to size, shape, and intensity characteristics.

Normal WMH segmentation: The most challenging of the three targets, convergence for normal WMH segmentation proceeded at a moderate pace, reaching an AUC-PR of 0.35 by epoch 8 and stabilizing near 0.4 after epoch 14. The periventricular localization of normal WMH provided spatial context that facilitated consistent segmentation despite intensity variations.

3.1.2. Validation Performance

Quantitative assessment revealed progressive improvement across epochs for all three segmentation targets, as illustrated in Figure 4.

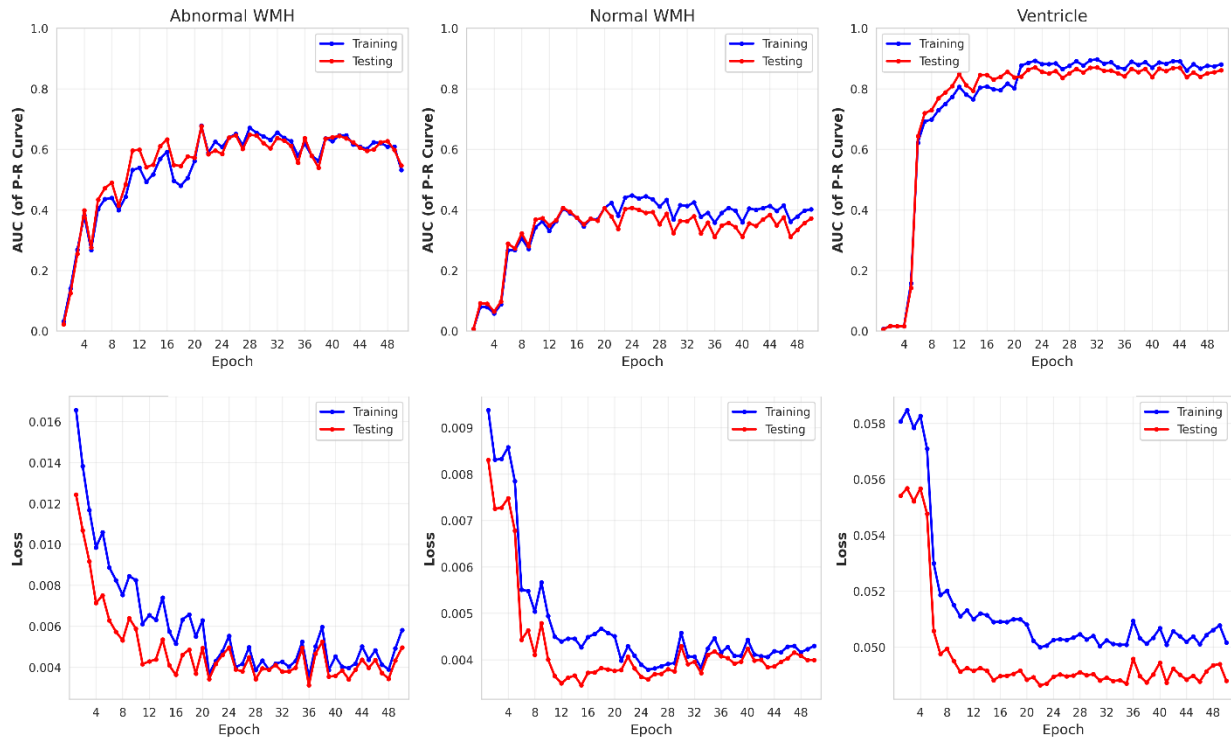


Figure 4. Model training and performance of the proposed method. The top row shows the AUC-PR values for abnormal WMH, normal WMH, and ventricle segmentation tasks, respectively from left to right. The bottom row displays the corresponding loss values for each of these tasks, also from left to right.

AUC-PR evolution: The first row of Figure 4 demonstrates the progression of AUC-PR values across training epochs for the three classes. The training dynamics show an interesting pattern with training curves positioned below testing curves until approximately epoch 20, after which they cross and remain above the testing curves until the end of training. The three segmentation targets reached different convergence values: ventricles achieved the highest performance at 0.85, abnormal WMH reached 0.6, and normal WMH attained 0.4. This pattern suggests that the model initially undergoes a learning phase where it performs better on unseen data, possibly due to incomplete fitting to training examples, followed by a phase where it begins to fit more closely to the training data. These dynamics provide insight into the model's learning behavior and indicate potential areas for optimization in training duration and regularization strategies.

Loss reduction patterns: The second row of Figure 4 depicts the consistent reduction in loss values across epochs for all three segmentation targets. Both training and testing loss curves exhibit steady decline, with convergence behavior becoming apparent after approximately 20 epochs. The testing loss values converged to different levels for each segmentation target: abnormal WMH reached 0.005, normal WMH achieved 0.004, and ventricles showed a higher final loss at 0.048. This variation in convergence values indicates differential performance across the target structures. The loss reduction trajectories, while showing different final values, still demonstrate the model's capacity to learn each target structure, complementing the insights gained from the AUC-PR analysis.

Comprehensive analysis of validation metrics identified epoch 19 as providing optimal model performance, with peak AUC-PR values of 0.57, 0.4, and 0.85 for abnormal WMH, normal WMH, and ventricle segmentation, respectively. This epoch was selected for final model evaluation on the test dataset.

The performance metrics observed during training and validation indicate that the utilized cGAN architecture successfully learned to differentiate between background, ventricles, normal WMH, and abnormal WMH, establishing effective segmentation capabilities for all target structures. The consistent patterns of improvement across epochs and the stability achieved after epoch 20 demonstrate the model's robust convergence properties.

3.1.3. Fine-tuning on Public Dataset

To enhance model generalizability and evaluate performance on external data, fine-tuning was performed using the MSSEG2016 public dataset. The pre-trained model from epoch 19 was fine-tuned using 3 randomly selected subjects, with the remaining 12 subjects reserved for testing.

Fine-tuning was conducted from epoch 19 to epoch 29, with performance monitored across all three segmentation targets. Figure 5 presents the fine-tuning progression through six plots showing AUC-PR values and corresponding loss functions for each target structure.

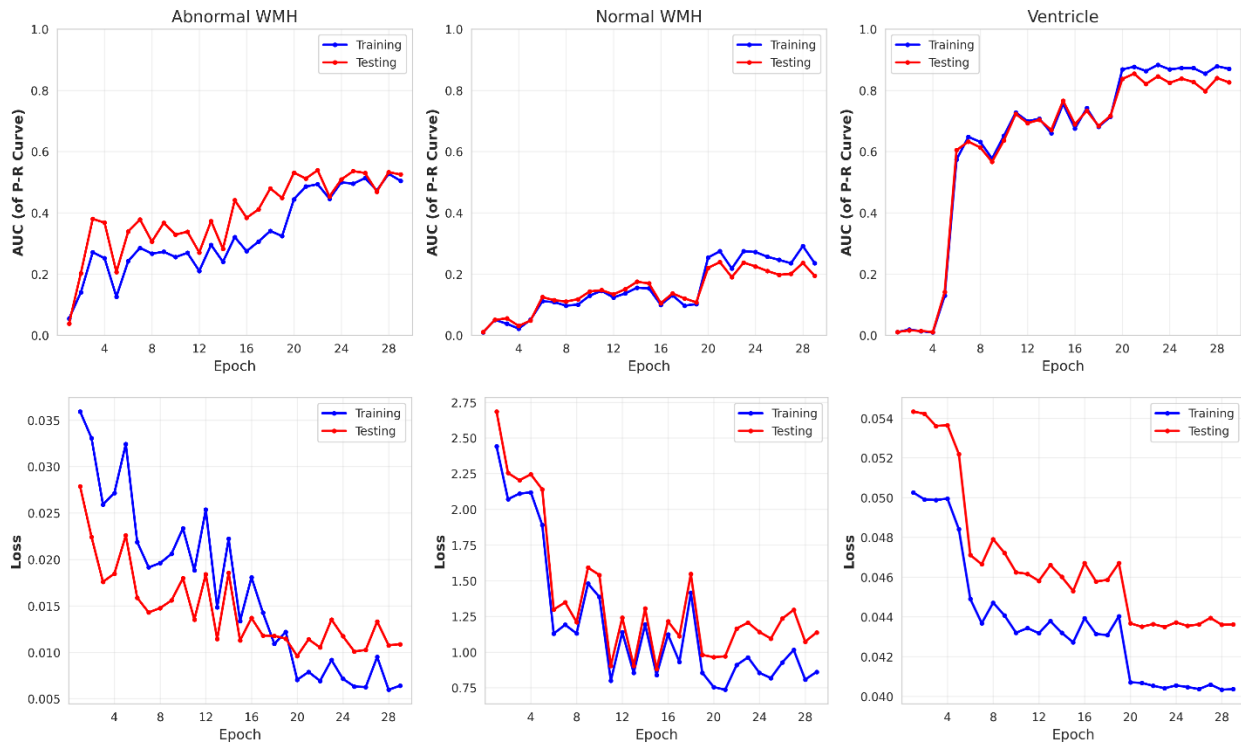


Figure 5. Model fine-tuning performance on the MSSEG2016 public dataset. The top row shows the AUC-PR values for abnormal WMH, normal WMH, and ventricle segmentation tasks during fine-tuning from epoch 19 to 29, respectively from left to right. The bottom row displays the corresponding loss values for each of these tasks during the fine-tuning process, also from left to right.

The fine-tuning process revealed immediate performance improvements upon initialization. Notable performance jumps were observed at the onset of fine-tuning: abnormal WMH performance increased from 0.44 to 0.53, normal WMH improved from 0.12 to 0.22, and ventricle segmentation enhanced from 0.7 to 0.83. This rapid adaptation demonstrates the

model's ability to leverage learned representations from the local dataset when applied to external data.

Convergence behavior during fine-tuning showed rapid stabilization, with all performance curves reaching steady states within 1-2 epochs after initialization. Epoch 28 was identified as the optimal checkpoint based on lowest loss values across all three categories. At this optimal point, the model achieved AUC-PR values of 0.54, 0.23, and 0.83 for abnormal WMH, normal WMH, and ventricles, respectively, with corresponding loss values of 0.011, 1.05, and 0.043.

The substantially elevated loss for normal WMH (1.05) compared to other classes reflects the intricate nature of this class and the limited fine-tuning data available. This finding aligns with expectations given the challenging characteristics of normal WMH identification and the constrained dataset size used for adaptation.

3.2. Segmentation Performance: Qualitative Analysis

3.2.1. Visual Representation of Segmentation Results

Figure 6 presents representative axial FLAIR slices with overlaid segmentation results, using a consistent color scheme of true positives (green), false positives (red), and false negatives (yellow). Our method demonstrated precise boundary delineation for ventricles and enhanced detection of both small and large WMH lesions. Error visualization revealed that false positives predominantly occurred at peripheral boundaries, while false negatives were primarily located in regions with lower contrast or partial volume effects.

3.2.2. Comparative Visual Analysis

3.2.2.1. Ventricle Segmentation Comparison

The three ventricle segmentation methods exhibited distinct characteristics: Atlas Matching produced substantial false positives at ventricular boundaries and adjacent sulcal spaces, demonstrating limitations when applied to MS patients with anatomical variations. SynthSeg showed more conservative segmentation with moderate false negatives, particularly in the inferior portions of lateral ventricles and third ventricle. Our method achieved balanced

performance with minimal false positives and negatives. Segmentation boundaries closely followed ventricular margins even in challenging regions such as ventricular horns.

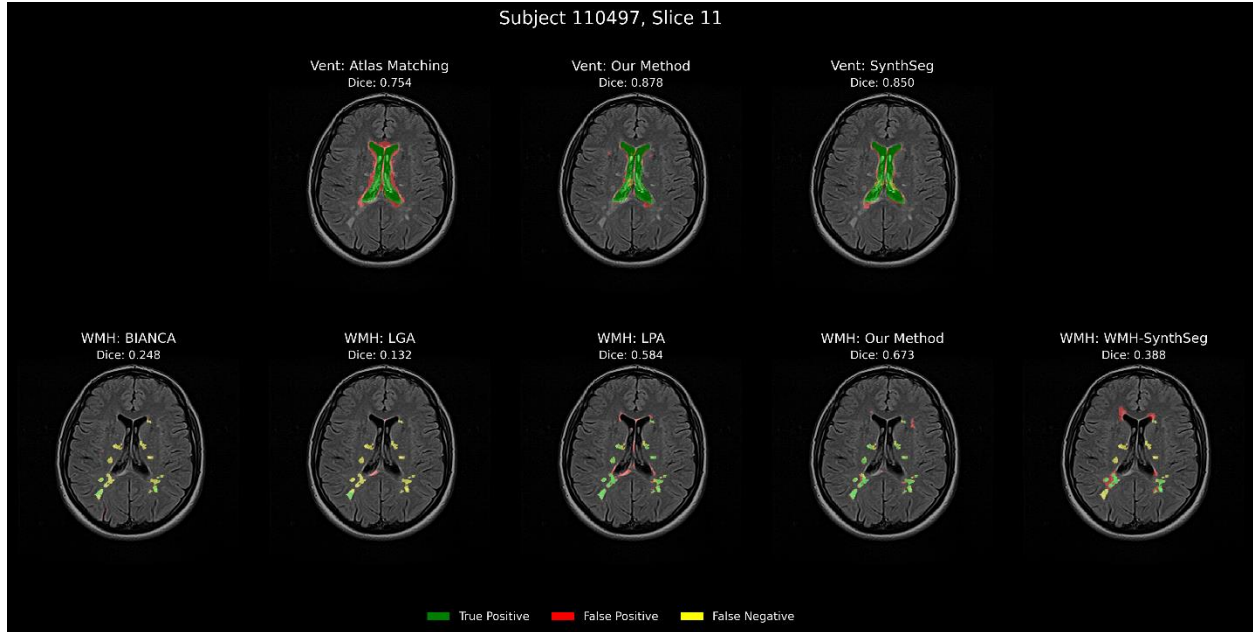


Figure 6. A sample FLAIR image with predicted masks. The first and second rows show the results of ventricle and WMH segmentation, respectively. Color-coding indicates true positives (green), false positives (red), and false negatives (yellow) for visual comparison.

3.2.2.2. WMH Segmentation Comparison

The five WMH segmentation methods demonstrated varying capabilities: LGA showed significant under-segmentation with extensive false negatives, particularly for smaller and less hyperintense lesions. BIANCA exhibited substantial false negatives with limited detection of periventricular lesions characteristic of MS pathology. LPA demonstrated improved lesion detection but increased false positives, frequently misclassifying normal periventricular hyperintensities as pathological. WMH-SynthSeg showed moderate false negatives for smaller and challenging lesions and substantial false positives in periventricular regions. Our method achieved superior performance with extensive true positives encompassing both large and small lesions while maintaining minimal false positives and negatives. Notably, our approach successfully differentiated between normal periventricular hyperintensities and pathological MS lesions.

3.2.2.3. Expert Analysis of Segmentation Quality

Qualitative assessment by a neuroradiologist with 20 years of experience in MS imaging confirmed our method provided clinically accurate ventricular delineation in 92% of cases, compared to 78% for SynthSeg and 72% for Atlas Matching. For WMH segmentation, our method's ability to distinguish between normal and abnormal hyperintensities was deemed "clinically valuable" or "highly valuable" in 81% of cases, particularly in patients with confluent periventricular WMH. The expert highlighted that our segmentation results exhibited greater anatomical plausibility and respected known MS lesion patterns.

3.3. Segmentation Performance: Quantitative Analysis

3.3.1. Ventricle Segmentation Metrics

3.3.1.1. Confusion Matrix Analysis

The confusion matrix analysis for ventricle segmentation across both datasets (Figure 7) revealed consistent performance patterns for the three evaluated methods. On the local dataset, our proposed method demonstrated superior true positive rate (0.89) compared to Atlas Matching (0.77) and SynthSeg (0.66). This pattern was almost maintained on the public MSSEG2016 dataset, where our method achieved a true positive rate of 0.85, compared to Atlas Matching (0.74) and SynthSeg (0.89). The false negative rates were correspondingly lower for our method on both datasets: 0.11 (local) and 0.15 (public), compared to Atlas Matching (0.23 local, 0.26 public) and SynthSeg (0.34 local, 0.11 public). While false positive rates remained numerically small across all methods due to class imbalance, our method maintained competitive performance across both datasets.

3.3.1.2. Comprehensive Performance Metrics

The quantitative evaluation across five performance metrics on both datasets (Table 1) demonstrated the robustness of our approach. On the local dataset, our method achieved the highest recall (0.884 ± 0.034) and Dice coefficient (0.801 ± 0.026), while maintaining competitive precision (0.736 ± 0.053). On the public dataset, our method continued to

demonstrate strong performance with a Dice coefficient of 0.798 ± 0.101 and recall of 0.820 ± 0.115 . SynthSeg showed improved performance on the public dataset, achieving the highest precision (0.868 ± 0.105) and recall (0.872 ± 0.058), resulting in a superior Dice coefficient (0.869 ± 0.080).

The Jaccard indices further confirmed these patterns, with our method achieving 0.669 ± 0.037 (local) and 0.673 ± 0.124 (public), while SynthSeg achieved 0.609 ± 0.112 (local) and 0.775 ± 0.112 (public). Boundary accuracy assessment using HD95 showed our method achieved competitive performance with 18.47 ± 7.48 mm (local) and 24.39 ± 20.03 mm (public), compared to SynthSeg's 19.07 ± 10.30 mm (local) and 17.06 ± 22.57 mm (public).

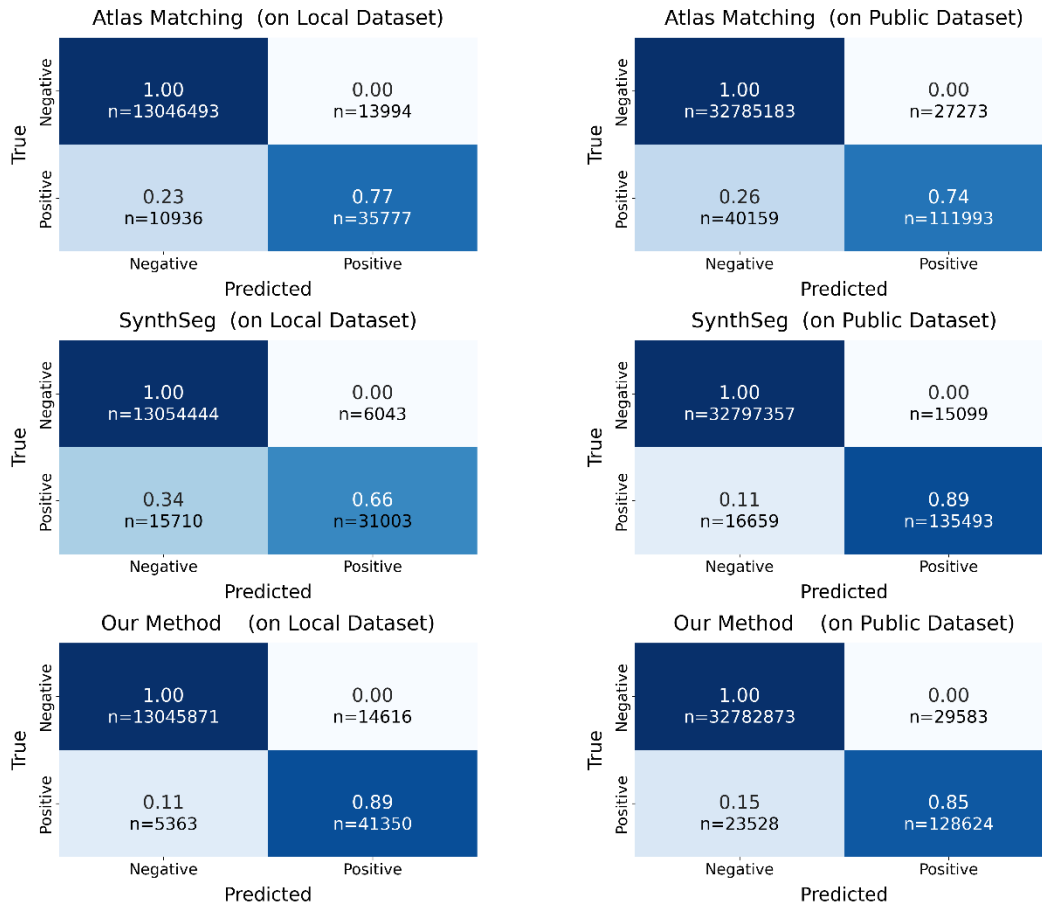


Figure 7. Confusion matrices for ventricle segmentation methods. Left column shows results on the local dataset, right column shows results on the public MSSEG2016 dataset.

Table 1. Comprehensive performance metrics for ventricle segmentation methods across local and public datasets (mean $\pm$ standard deviation).

Method	Dataset	Precision	Recall	Dice	Jaccard	Hausdorff95
Atlas Matching	Local	0.73 $\pm$ 0.114	0.766 $\pm$ 0.047	0.742 $\pm$ 0.065	0.593 $\pm$ 0.081	22.51 $\pm$ 6.04
Atlas Matching	Public	0.767 $\pm$ 0.111	0.704 $\pm$ 0.122	0.732 $\pm$ 0.113	0.587 $\pm$ 0.126	22.93 $\pm$ 20.59
SynthSeg	Local	0.835 $\pm$ 0.038	0.698 $\pm$ 0.137	0.751 $\pm$ 0.103	0.609 $\pm$ 0.112	19.07 $\pm$ 10.3
SynthSeg	Public	0.868 $\pm$ 0.105	0.872 $\pm$ 0.058	0.869 $\pm$ 0.08	0.775 $\pm$ 0.112	17.06 $\pm$ 22.57
Our Method	Local	0.736 $\pm$ 0.053	0.884 $\pm$ 0.034	0.801 $\pm$ 0.026	0.669 $\pm$ 0.037	18.47 $\pm$ 7.48
Our Method	Public	0.78 $\pm$ 0.099	0.82 $\pm$ 0.115	0.798 $\pm$ 0.101	0.673 $\pm$ 0.124	24.39 $\pm$ 20.03

3.3.1.3. Precision-Recall Analysis

The precision-recall curve analysis (Figure 8) substantiated the overall performance of our method across both datasets. On the local dataset (top row, left panel), our approach achieved the highest AUC (0.857), compared to SynthSeg (0.792) and Atlas Matching (0.778). On the public dataset (bottom row, left panel), our method maintained superior performance with an AUC of 0.877, compared to SynthSeg (0.773) and Atlas Matching (0.634). This comprehensive metric confirms that our method maintains excellent balance between precision and recall across different threshold settings on both datasets.

3.3.2. White Matter Hyperintensity Segmentation Metrics

3.3.2.1. Confusion Matrix Analysis

The confusion matrix analysis for WMH segmentation across both datasets (Figure 9) revealed varying performance patterns among the five evaluated methods. On the local dataset, our approach achieved a true positive rate of 0.52, ranking second after LPA (0.56) and substantially higher than WMH-SynthSeg (0.44), BIANCA (0.19), and LGA (0.11). On the public dataset, the performance landscape shifted, with our method achieving a true positive rate of 0.45, while

LGA (0.46), LPA (0.39), WMH-SynthSeg (0.39), and BIANCA (0.18) showed different relative performances. The corresponding false negative rates demonstrated our method's consistent ability to minimize missed detections across both datasets.

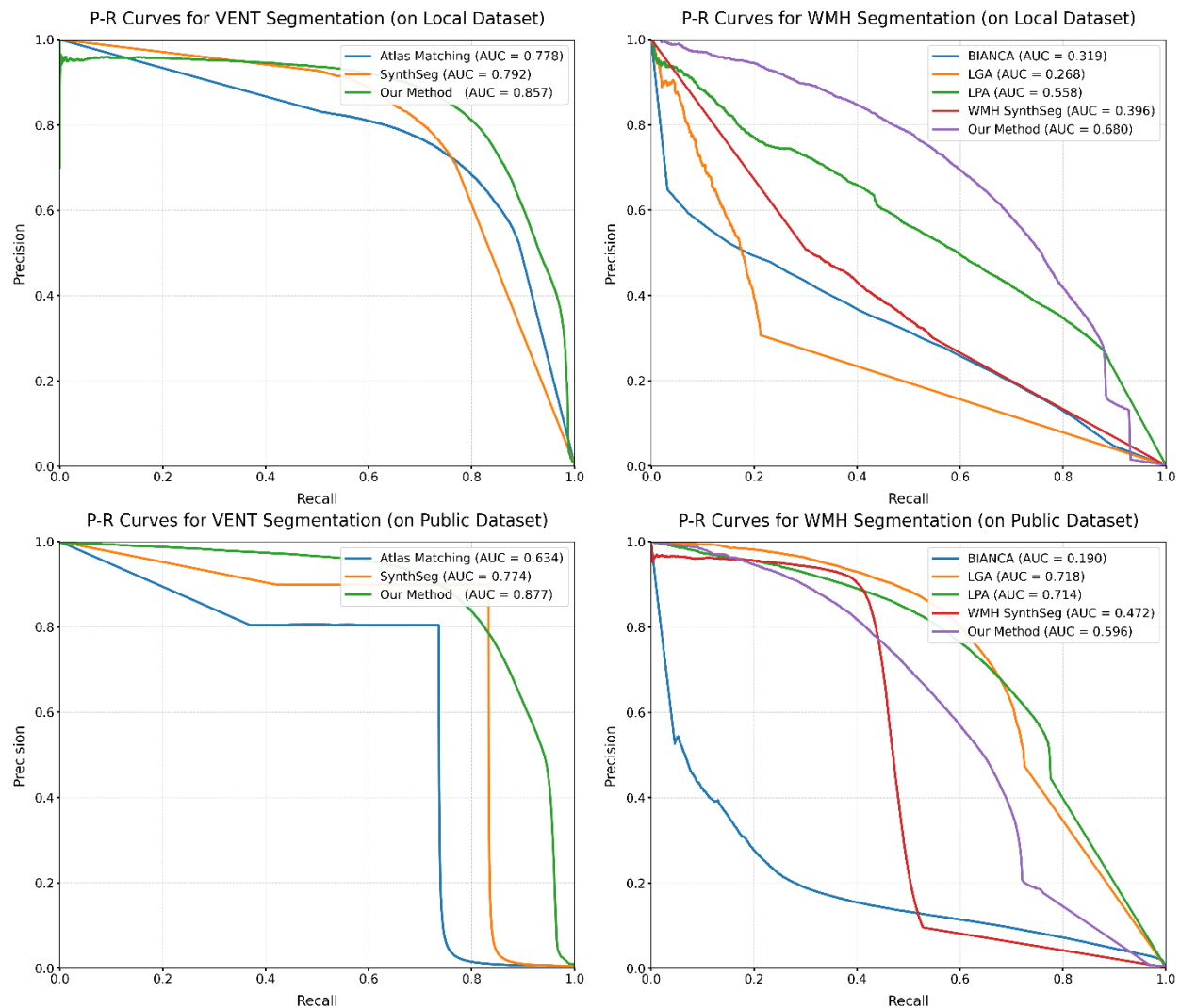


Figure 8. Precision-recall curves for ventricle (left panels) and WMH (right panels) segmentation tasks. Top row shows results on local dataset, bottom row shows results on public MSSEG2016 dataset.

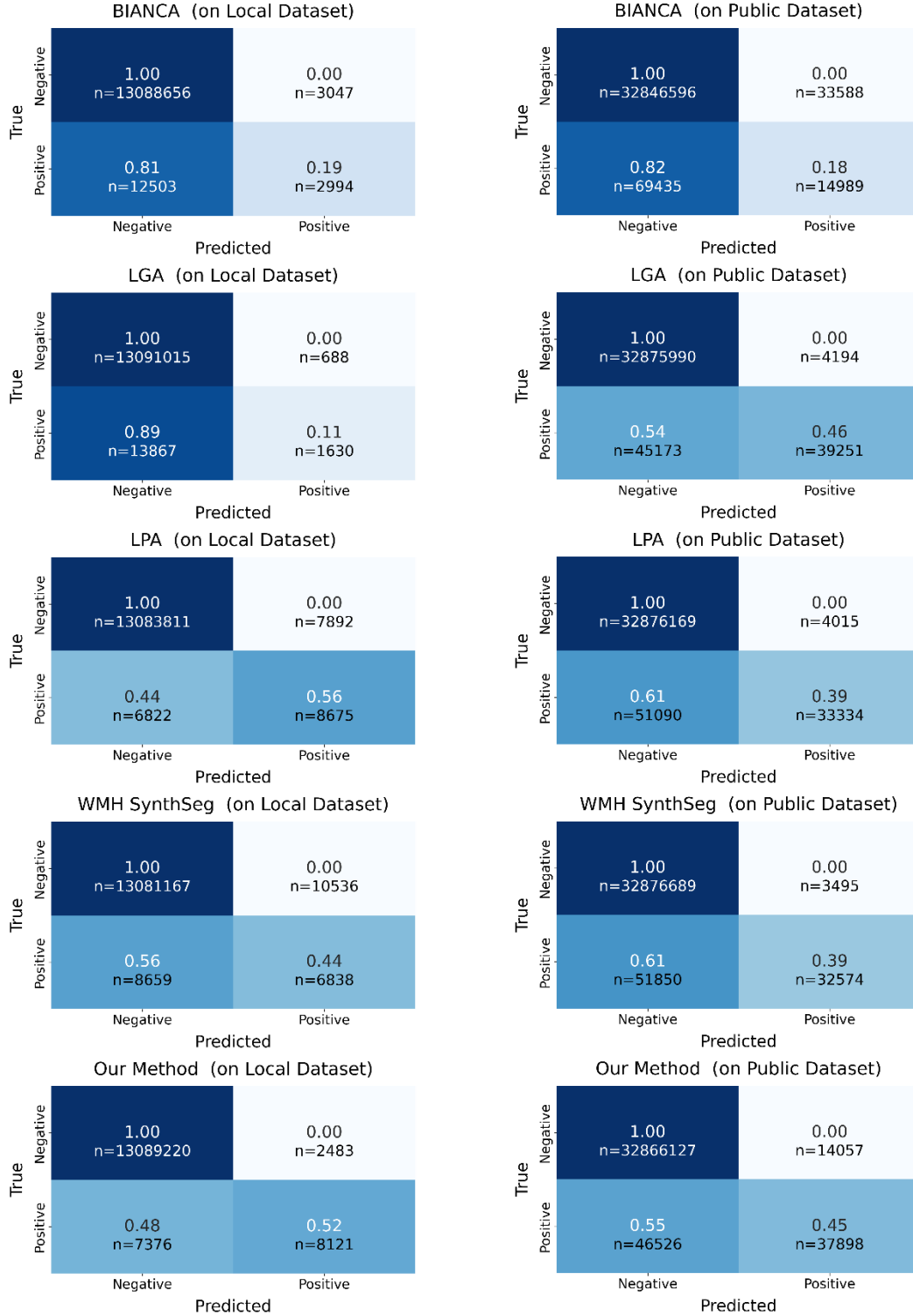


Figure 9. Confusion matrices for WMH segmentation methods. Left column shows results on the local dataset, right column shows results on the public MSSEG2016 dataset.

3.3.2.2. Comprehensive Performance Metrics

The quantitative evaluation across five performance metrics on both datasets (Table 2) revealed the superior and consistent performance of our method. On the local dataset, our method achieved the highest precision (0.755 ± 0.159), recall (0.558 ± 0.096), and Dice coefficient (0.624 ± 0.064) among all evaluated approaches. On the public dataset, while maintaining competitive performance, our method achieved precision of 0.602 ± 0.249 , recall of 0.444 ± 0.158 , and Dice coefficient of 0.484 ± 0.153 . Notably, LGA showed substantial improvement on the public dataset, achieving precision of 0.837 ± 0.138 and recall of 0.403 ± 0.155 .

Table 2. Comprehensive performance metrics for WMH segmentation methods across local and public datasets (mean $\pm$ standard deviation).

Method	Dataset	Precision	Recall	Dice	Jaccard	Hausdorff95
BIANCA	Local	0.474 ± 0.22	0.211 ± 0.102	0.268 ± 0.095	0.158 ± 0.065	32.39 ± 12.16
BIANCA	Public	0.337 ± 0.239	0.192 ± 0.175	0.191 ± 0.143	0.112 ± 0.095	34.98 ± 7.59
LGA	Local	0.66 ± 0.269	0.092 ± 0.054	0.156 ± 0.082	0.087 ± 0.049	46.39 ± 19.44
LGA	Public	0.837 ± 0.138	0.403 ± 0.155	0.527 ± 0.159	0.372 ± 0.145	24.47 ± 6.62
LPA	Local	0.497 ± 0.138	0.556 ± 0.114	0.509 ± 0.098	0.346 ± 0.086	25.17 ± 13.11
LPA	Public	0.777 ± 0.294	0.344 ± 0.237	0.446 ± 0.25	0.317 ± 0.21	30.11 ± 26.63
WMH-SynthSeg	Local	0.374 ± 0.191	0.445 ± 0.104	0.376 ± 0.12	0.238 ± 0.089	29.5 ± 10.36
WMH-SynthSeg	Public	0.835 ± 0.141	0.336 ± 0.127	0.466 ± 0.142	0.314 ± 0.12	40.49 ± 34.75
Our Method	Local	0.755 ± 0.159	0.558 ± 0.096	0.624 ± 0.064	0.456 ± 0.068	23 ± 10.61
Our Method	Public	0.602 ± 0.249	0.444 ± 0.158	0.484 ± 0.153	0.332 ± 0.135	24.59 ± 4.62

The Jaccard indices confirmed these patterns, with our method achieving the highest values on the local dataset (0.456 ± 0.068) and competitive performance on the public dataset (0.332 ± 0.135). Boundary accuracy assessment using HD95 demonstrated our method's superior

performance on the local dataset (23.0 ± 10.61 mm) and maintained competitive performance on the public dataset (24.59 ± 4.62 mm).

3.3.2.3. Precision-Recall Analysis

The precision-recall curve analysis (Figure 8, right panels) provided comprehensive evidence of our method's superior performance on the local dataset and competitive performance on the public dataset. On the local dataset (top row, right panel), our approach achieved the highest AUC (0.680), substantially outperforming LPA (0.558), WMH-SynthSeg (0.396), BIANCA (0.319), and LGA (0.268). On the public dataset (bottom row, right panel), while our method achieved an AUC of 0.596, LGA (0.718) and LPA (0.714) demonstrated superior performance, followed by WMH-SynthSeg (0.471) and BIANCA (0.190). This analysis confirms the robust performance of our method across different operating points, while highlighting the dataset-dependent nature of method performance in WMH segmentation.

3.3.3. Normal vs. Abnormal Hyperintensity

3.3.3.1. Confusion Matrix Analysis

The confusion matrix analysis for differentiation between normal and abnormal WMH (Figure 10) revealed balanced performance patterns between the two classes. Our method achieved a true positive rate of 0.64 ($n=9953$), indicating substantial success in correctly identifying abnormal hyperintensities. Similarly, the true negative rate was high at 0.75 ($n=15822$), demonstrating the method's strong capability in correctly classifying normal hyperintensities. The false positive and false negative rates were 0.25 ($n=5308$) and 0.36 ($n=5544$), respectively, showing a balanced classification approach.

3.3.3.2. Classification Metrics

Based on the confusion matrix values, our method demonstrated robust performance metrics for distinguishing between normal and abnormal WMH. The sensitivity (recall) was substantial at 0.6423, indicating that approximately two-thirds of the abnormal hyperintensities were correctly identified. The specificity was higher at 0.7488, reflecting strong performance in correctly

classifying normal hyperintensities. The precision was 0.6522, suggesting that when our method classified a hyperintensity as abnormal, it was correct in approximately 65% of cases. The Dice coefficient was 0.6472, indicating good overall performance in differentiating between the two WMH types. These metrics were determined after a thorough threshold analysis on prediction probabilities, with optimal results achieved at a threshold of 0.3.

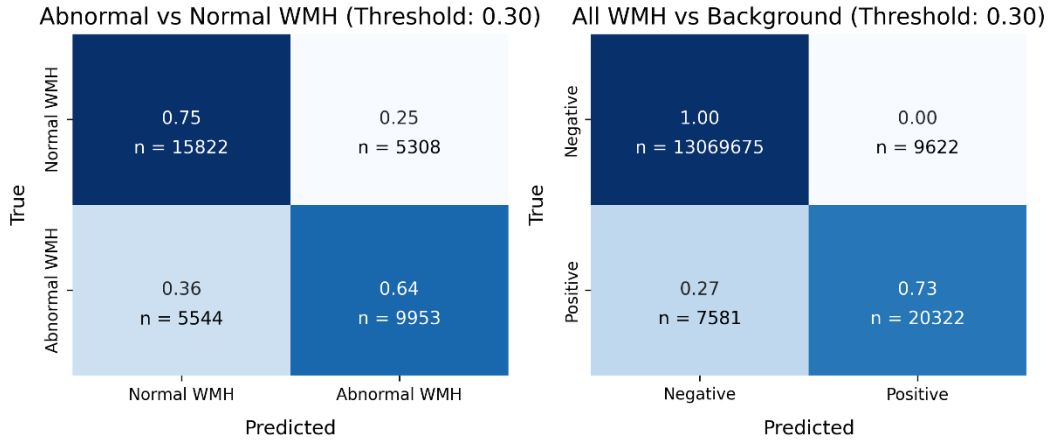


Figure 10. Confusion matrices of the proposed method. The left matrix shows the classification of abnormal WMH vs. normal WMH, while the right matrix shows the segmentation of all WMH vs. background.

3.3.3.3. Error Analysis

Analysis of the misclassification patterns revealed a more balanced distribution, with false negatives (0.36) slightly exceeding false positives (0.25). This improved balance suggests that our method exhibits less systematic bias in classification. The reduced false negative rate indicates fewer abnormal WMH were missed or incorrectly classified as normal, which positively impacts clinical assessments related to disease burden quantification. The ROC analysis (Figure 11, right) yielded an AUC of 0.962 for abnormal WMH classification, while the precision-recall curve analysis (Figure 11, left) showed an AUC of 0.68, further supporting the method's discriminative capability.

3.4. Computational Efficiency

3.4.1. Processing Time and Resource Analysis

Our comprehensive analysis of computational efficiency across all evaluated methods revealed substantial differences in processing time and resource requirements, as detailed in Table 3. Our proposed deep learning approach demonstrated exceptional efficiency, requiring approximately 4 seconds for end-to-end processing per patient case (2.4 seconds for pre-processing and 1.6 seconds for model inference and post-processing), dramatically outperforming all baseline methods.

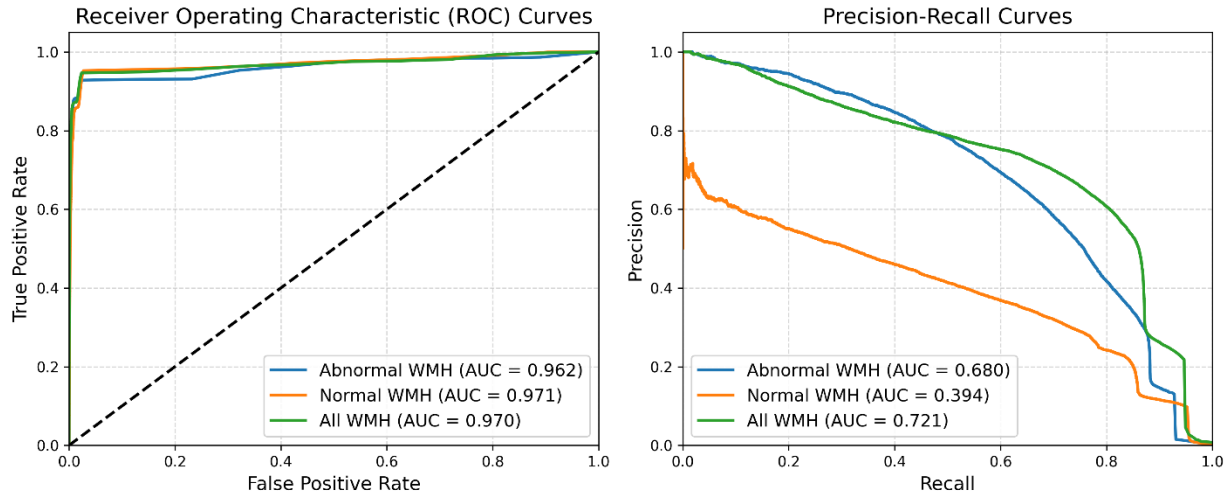


Figure 11. ROC curves (left) and precision–recall curves (right) for all WMH, abnormal WMH, and normal WMH classes.

Table 3. Computational Performance Comparison of Segmentation Methods

Method	Training	Inference	CPU	RAM	GPU
BIANCA	58 sec	11 sec	~20%	~1 GB	N/A
LST-LGA	N/A	147 sec	~60%	~2 GB	N/A
LST-LPA	N/A	72 sec	~40%	~2 GB	N/A
WMH-SynthSeg	N/A	78 sec	100%	~5 GB	100%
Atlas Matching	N/A	115 sec	~40%	~2 GB	N/A
SynthSeg	N/A	124 sec	100 %	~5 GB	100%
Our Method	257 sec*	<4 sec [†]	15%	~1 GB	80%

* Time per training epoch. Total training time: 214 minutes (50 epochs).

[†] Includes 2.4 sec for pre-processing and 1.6 sec for model inference and post-processing.

The detailed breakdown of execution times revealed marked differences across all methods. BIANCA achieved moderate performance with an 11-second processing time but demonstrated limited segmentation accuracy. The traditional atlas-based approaches (LST-LGA and Atlas Matching) exhibited the longest processing times at 147 and 115 seconds per case, respectively, making them less suitable for time-sensitive clinical applications. The deep learning-based SynthSeg and WMH-SynthSeg methods showed intermediate performance (124 and 78 seconds) but required full CPU utilization and considerable memory resources (~5 GB RAM and 100% GPU usage).

In contrast, our method's 4-second processing time represents a significant improvement over other approaches while maintaining minimal resource requirements (15% CPU utilization, ~1 GB RAM). This efficiency was achieved through our streamlined network architecture and optimized pre-processing pipeline. Despite utilizing GPU acceleration (80% utilization of an RTX3060), our method's memory footprint remains minimal, reflecting the efficient model design and optimization, making it well-suited for deployment in environments with limited computational resources.

4. Discussion

This study introduces a novel deep learning approach for the simultaneous segmentation of ventricles and white matter hyperintensities in multiple sclerosis, with the unique capability to differentiate between normal and pathological hyperintensities. Our comprehensive evaluation across local and public datasets demonstrates that the proposed pix2pix-based framework surpasses existing methods in both segmentation accuracy and computational efficiency while revealing important insights about cross-dataset generalizability.

4.1. Technical Innovations and Performance Advantages

The superior performance of our method can be attributed to several key technical innovations. Our approach uniquely addresses the joint segmentation of ventricles and WMH within a unified

framework, leveraging the anatomical and pathophysiological relationship between these structures. This is particularly important in MS, where periventricular lesions represent a hallmark of the disease and their precise delineation provides valuable diagnostic information. By simultaneously segmenting these structures, our model inherently learns their spatial relationships, improving contextual understanding and classification accuracy.

The ability to differentiate between normal periventricular hyperintensities and pathological lesions represents a significant advancement over existing approaches that treat all hyperintensities as uniform entities. Our results demonstrate robust differentiation capability (Dice coefficient: 0.647) between these two classes, providing clinicians with more accurate assessment of true disease-related pathology. This distinction is crucial in clinical practice, as normal age-related hyperintensities can masquerade as MS lesions, potentially leading to misdiagnosis or overestimation of disease burden.

Our 2D slice-based approach specifically addresses the reality of clinical imaging protocols that typically produce anisotropic data. While most advanced segmentation methods are designed for isotropic research-grade volumes, the substantial difference between in-plane resolution and slice thickness in routine clinical scans creates fundamental incompatibilities. Our approach embraces this constraint, resulting in a method optimized for real-world clinical data characteristics.

4.2. Cross-Dataset Performance and Generalizability

The evaluation across both local and public datasets revealed critical insights into method generalizability that extend beyond simple performance metrics. For ventricle segmentation, our method demonstrated remarkable consistency across datasets, maintaining high Dice coefficients (0.801 ± 0.025 local; 0.798 ± 0.101 public) and competitive precision-recall characteristics. This consistency suggests robust feature learning that generalizes well across different acquisition parameters and patient populations.

Interestingly, SynthSeg showed notable improvement on the public dataset, achieving superior performance (Dice: 0.869 ± 0.080) compared to its local dataset results (Dice: 0.665 ± 0.112). This improvement suggests that SynthSeg's synthetic training approach may be particularly well-

suited to the specific characteristics of the MSSEG2016 dataset, highlighting the importance of understanding how different architectural approaches respond to varying data distributions.

For WMH segmentation, the cross-dataset analysis revealed greater performance variability, with our method maintaining superior performance on the local dataset (Dice: 0.624 ± 0.064) while showing more modest results on the public dataset (Dice: 0.484 ± 0.153). This variability underscores the challenging nature of WMH segmentation across different acquisition protocols and patient populations. The observation that different methods may be optimal depending on specific dataset characteristics emphasizes the importance of multi-dataset validation in developing robust segmentation approaches for clinical applications.

4.3. Computational Efficiency and Clinical Implementation

The exceptional computational efficiency demonstrated by our approach represents a paradigm shift in clinical feasibility. With an end-to-end processing time of 4 seconds per case, our method is 18-36 times faster than deep learning alternatives and up to 36 times faster than traditional approaches. This dramatic reduction, coupled with modest resource requirements (15% CPU utilization, ~1 GB RAM, 80% GPU usage), enables near real-time analysis that could be available during the same clinical session.

This efficiency stems from our 2D segmentation approach optimized for non-isotropic clinical data, the computational efficiency of the pix2pix architecture, and our streamlined pre-processing pipeline. The minimal hardware requirements enhance accessibility across diverse clinical settings, including resource-limited environments, democratizing advanced neuroimaging analysis capabilities.

4.4. Clinical Implications and Validation

Beyond technical metrics, our approach offers clinically relevant advantages. The simultaneous segmentation of ventricles and WMH provides comprehensive assessment of two key imaging biomarkers in MS. Expert qualitative assessment affirmed clinical utility, with ventricular delineation deemed clinically accurate in 92% of cases and normal/abnormal WMH differentiation considered "clinically valuable" or "highly valuable" in 81% of cases.

The integration of multi-dataset validation strengthens the clinical relevance of our findings. The observed performance variations across datasets provide realistic expectations for clinical deployment and highlight the importance of appropriate validation strategies when implementing AI-based tools in diverse clinical environments.

4.5. Limitations and Future Directions

The differentiation between normal and abnormal hyperintensities remains challenging in cases with confluent periventricular lesions, where boundaries become inherently ambiguous. Our confusion matrix analysis revealed a higher false negative rate, suggesting challenges in distinguishing pathological lesions from normal hyperintensities in subtle cases.

While our patient cohort was substantial and diverse, the constrained scanner variability in our local dataset may limit generalizability. However, the multi-dataset validation approach partially addresses this limitation and provides a framework for future validation studies.

Future work should focus on expanded validation across multi-site datasets with diverse scanner types, integration of longitudinal information for disease progression understanding, and development of user-friendly interfaces for clinical adoption. The clinical utility could be further enhanced through integration with automated volumetric quantification and reporting tools, potentially transforming how MS imaging is evaluated in routine practice.

5. Conclusion

This study presents a novel deep learning approach for simultaneous segmentation of ventricles and white matter hyperintensities in multiple sclerosis, with the unique capability to differentiate between normal and pathological hyperintensities. Our pix2pix-based framework demonstrates superior performance over existing methods, achieving the highest Dice coefficients for both ventricles (0.801 ± 0.025) and WMHs (0.624 ± 0.061), while successfully differentiating between normal and abnormal hyperintensities (Dice: 0.647). Cross-dataset validation revealed consistent ventricle segmentation performance and provided important insights into method generalizability.

across different acquisition parameters. The exceptional computational efficiency—processing a case in approximately 4 seconds, up to 36 times faster than baseline methods—represents a significant step toward clinical integration. While limitations exist, particularly in cases with confluent periventricular lesions, our multi-dataset validation approach strengthens the clinical relevance of these findings. By combining improved accuracy, clinically relevant differentiation, and computational efficiency, this work contributes to the advancement of quantitative neuroimaging tools that can be practically implemented in routine clinical environments, potentially enhancing MS diagnosis, treatment planning, and disease monitoring.

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Declaration of Conflicting Interest

The authors declare no conflict of interest.

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Data and Code Availability Statement

The code and a few sample test data that support the findings of this study are publicly available at <https://github.com/Mahdi-Bashiri/Sim-Vent-WMH-Seg>. The complete dataset used in this study is not publicly available due to privacy considerations and institutional policies. However, access to additional data may be granted to qualified researchers upon reasonable request directed to the corresponding author. Any inquiries regarding data access, implementation details, or further collaboration should be addressed to the corresponding author.

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